

1.6 Electrochemistry

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.6 Electrochemistry
Difficulty	Easy

Time allowed: 40

Score: /27

Percentage: /100

Question 1a

This question is about oxidation numbers.

Complete Table 1.1 giving the oxidation number of sulfur in the following compounds.

Table 1.1

Compound	Oxidation number
SO ₂	
SO ₃	
H ₂ SO ₄	
SF ₄	

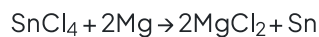
[4 marks]**Question 1b**

State the oxidation number of sulfur in its elemental form of S₈.

[1 mark]

Question 1c

The reaction between tin chloride and magnesium is classed as a redox reaction.



i)

State the change in oxidation number of magnesium.

[1]

ii)

State the change in oxidation number of tin.

[1]

iii)

Deduce which species is acting as the reducing agent.

[1]

[3 marks]

Question 1d

Antimony reacts with nitric acid to form antimony oxide, nitrogen monoxide and water as shown the following equation.



i)

State the change in oxidation number of antimony.

[1]

ii)

State the change in oxidation number of nitrogen.

[1]

iii)

Identify the oxidising agent in this reaction.

[1]

[3 marks]

Question 2aState **two** definitions of reduction.

[2 marks]

Question 2b

Complete Table 2.1 with the oxidation numbers of the elements in the following species.

Table 2.1

Species	Oxidation number
S^{2-}	
Pb^{2+}	
V^{3+}	
Si	
Sb^{3+}	
H^-	

[3 marks]

Question 2c

Oxidation number are sometimes visible in the names of chemicals.

Complete Table 2.2 with the oxidation numbers of the given elements in the following species.

Table 2.2

Compound	Oxidation number
Copper in copper(I) oxide	
Iron in iron(III) chloride	
Tin in tin(IV) chloride	
Vanadium in vanadium(II) oxide	

[3 marks]

Question 3a

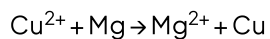
This question is about disproportionation reactions.

Explain what is meant by the term *disproportionation reaction*.

[2 marks]

Question 3b

State whether the displacement reaction between copper(II) ions and magnesium is an example of a disproportionation reaction. Explain your answer.



[2 marks]

Question 3c

i)

Write a balanced equation for the disproportionation reaction of sulfuric acid with copper(I) oxide to form copper, copper sulfate and water.

[1]

ii)

Identify which element, in this reaction, undergoes disproportionation.

[1]

[2 marks]**Question 3d**

i)

Write a balanced equation for the reaction of nitrogen dioxide with water to form nitric acid and nitrous acid, HNO_2 .

[1]

ii)

Identify which element undergoes disproportionation.

[1]

[2 marks]

Model Answers: Easy

1a

a) The oxidation number of sulfur in the following compounds are:

Compound	Oxidation number
SO ₂	+4 ; [1 mark]
SO ₃	+6 ; [1 mark]
H ₂ SO ₄	+6 ; [1 mark]
SF ₄	+4 ; [1 mark]

[Total: 4 marks]

- In order to determine the oxidation number, you need to know the rules!
- For compounds that contain hydrogen and oxygen, hydrogen is always +1 (except in hydrides) and oxygen is -2 (except in OF₂)
- If the compound does not have a charge, then the oxidation numbers must add up to 0
- Therefore
 - SO₂ = S is +4 as there are two oxygen atoms with an oxidation number of -2
 - SO₃ = S is +6 as there are three oxygen atoms with an oxidation number of -2
 - H₂SO₄ = S is +6 as there are four oxygen atoms with an oxidation number of -2 and two hydrogen atoms with a charge of 2+
 - S in H₂SO₄ = (+2) + (-8) = -6, so S must be +6 to equal 0
 - SF₄ = each F atom has a charge of -1, therefore sulfur must have an oxidation number of +4 to equal 0

1b

b) The oxidation number of sulfur in S₈ is:

- Zero / 0; [1 mark]

[Total: 1 mark]

- Elements have an oxidation number of zero

1c

c)

i) The change in oxidation number of magnesium is:

- 0 to +2; [1 mark]

ii) The change in oxidation number of tin is:

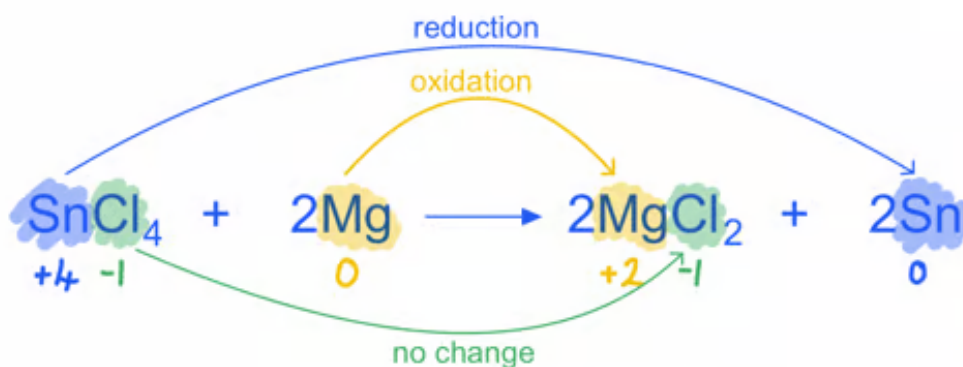
- +4 to 0; [1 mark]

iii) The reducing agent is:

- Magnesium / Mg; [1 mark]

[Total: 3 marks]

- This question is more likely to be asked as 3 mark question to determine the oxidising or reducing agent
- In order to gain the marks, you will need to give the oxidation numbers and state the change in oxidation number
- It can be helpful to write out the equation and connect the products and reactants – as shown below



- If the oxidation number increases, oxidation has occurred (electrons have been lost)

- The reducing agent is itself oxidised (Mg)
- If the oxidation number decreases, reduction has occurred (electrons have been gained)
 - The oxidising agent is itself reduced (SnCl_4)

1d

d)

i) The change in oxidation number of antimony is:

- 0 to +3; [1 mark]

ii) The change in oxidation number of nitrogen is:

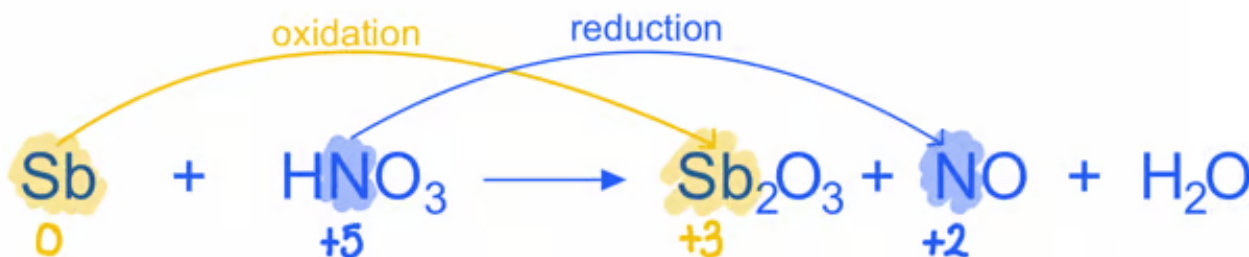
- +5 to +2; [1 mark]

iii) The oxidising agent is:

- Nitric acid / HNO_3 ; [1 mark]

[Total: 3 marks]

- We can use the same method as used in part (c) to determine the oxidising agent



- Oxidising agents cause oxidation, which means that they are themselves reduced
 - Since nitric acid is reduced, it must be the oxidising agent

2a

a) **Two** definitions of reduction are:

Any **two** of the following:

- The loss of oxygen; [1 mark]
- The addition / gain of electrons; [1 mark]
- A decrease in oxidation number; [1 mark]
- The addition / gain of hydrogen; [1 mark]

[Total: 2 marks]

- You should know the first three definitions of reduction and oxidation
 - The definitions involving hydrogen are also useful to know
- If you are asked to identify if a reaction is reduction or oxidation, you should start by looking for:
 - Any gain / loss of oxygen
 - Then, any gain / loss of hydrogen
 - This can save you the time and effort
 - Finally, check electrons and oxidation numbers

2b

b) The oxidation numbers are:

Species	Oxidation number
S^{2-}	-2
Pb^{2+}	+2
V^{3+}	+3
Si	0
Sb^{3+}	+3
H^-	-1

- All oxidation states correct; [3 marks]
- 3 - 5 oxidations numbers correct; [2 marks]
- 1 - 2 oxidations numbers correct; [1 mark]

[Total: 3 marks]

- The oxidation state of a simple ion is the charge on the ion
 - Group 1 metal ions, e.g. K^+ have an oxidation number of +1
 - Group 2 metal ions, e.g. Ca^{2+} have an oxidation number of +2
 - Halide ions, e.g. Br^- have an oxidation number of -1
 - Group 16 non-metal ions, e.g. S^{2-} have an oxidation number of -2

- Transition metal ions have variable oxidation numbers, so you need to look at the charge, e.g. vanadium has four common oxidation numbers: +2, +3, +4, +5
- **Careful:** Out of habit you might put the oxidation number of hydrogen in this question as +1 **but** it is not a standard hydrogen ion / H^+ , it is the hydride ion / H^- which has an oxidation number of -1

2c

c) The completed table is:

Compound	Oxidation number
Cu in copper(I) oxide	+1
Iron in iron(III) chloride	+3
Tin in tin(IV) chloride	+4
Vanadium in vanadium(II) oxide	+2

- All 4 correct; [3 marks]
- 2 or 3 correct ; [2 marks]
- 1 correct ; [1 mark]

[Total: 3 marks]

- Some of the compounds in the question are known by other more common names
 - Copper(I) oxide is cuprous oxide
- If you are asked about the oxidation number in an exam and you see Roman numerals, your answer is right there

2c

c) The completed table is:

Compound	Oxidation number
Cu in copper(I) oxide	+1
Iron in iron(III) chloride	+3
Tin in tin(IV) chloride	+4
Vanadium in vanadium(II) oxide	+2

- All 4 correct; [3 marks]
- 2 or 3 correct ; [2 marks]
- 1 correct ; [1 mark]

[Total: 3 marks]

- Some of the compounds in the question are known by other more common names
 - Copper(I) oxide is cuprous oxide
- If you are asked about the oxidation number in an exam and you see Roman numerals, your answer is right there

3a

a) A disproportionation reaction is:

- (A reaction) where the same species undergoes reduction and oxidation; [1 mark]
- Simultaneously / at the same time; [1 mark]

[Total: 2 marks]

- The word disproportionate literally means not (dis) balanced (proportionate)
- Applying this idea to a single chemical suggests that there are two things happening to the chemical
 - This would be reduction and oxidation at the same time

3b

b) The displacement reaction between copper(II) ions and magnesium:

b) The displacement reaction between copper(II) ions and magnesium:

- Is not a disproportionation reaction; [1 mark]
- (Because) reduction and oxidation do not happen to one / the same species

OR

(Because) the Cu^{2+} is reduced while the Mg is oxidised; [1 mark]

[Total: 2 marks]

- **Remember:** Disproportionation is where the same species undergoes reduction and oxidation at the same time
 - In this question, one species is being oxidised and another species is being reduced so it cannot be disproportionation
- **Careful:** The word displacement is there to distract you – do not automatically think that because it is a **DIS**placement reaction, it is a **DIS**proportionation reaction as well

3c

i) The balanced equation for the reaction of copper(I) oxide with sulfuric acid to form copper, copper sulfate and water is:

- $\text{H}_2\text{SO}_4 + \text{Cu}_2\text{O} \rightarrow \text{Cu} + \text{CuSO}_4 + \text{H}_2\text{O}$; [1 mark]

ii) The element which undergoes disproportionation is:

- Copper / Cu; [1 mark]

[Total: 2 marks]

- This question is asking you to write a balanced symbol equation **but** for one mark, there may not any balancing to do
 - You are expected to determine the formula of copper(I) oxide from its name
 - You are expected to know the formulae for the remaining chemicals
- The only element that could undergo disproportionation is copper
 - Hydrogen is only involved in **one** product
 - Sulfur is only involved in **one** product
 - Oxygen is involved in **two** reactants
- Copper is +1 in copper(I) oxide, 0 in copper and +2 in copper sulfate
 - It has been reduced from +1 in copper(I) oxide to 0 in copper
 - It has been oxidised from +1 in copper(I) oxide to +2 in copper sulfate
 - Therefore copper has undergone disproportionation

- therefore copper has undergone disproportionation.

3d

d)

i) The balanced equation for the reaction of nitrogen dioxide with water to form nitric acid and nitrous acid, HNO_2 , is:



ii) The element which undergoes disproportionation is:

- Nitrogen / N; [1 mark]

[Total: 2 marks]

- You are given the formula for nitrous acid
- You are expected to know the formulae for the remaining chemicals
- The only element that could undergo disproportionation is nitrogen
 - Oxygen is involved in **two** reactants
 - Hydrogen is always +1 unless it is in a metal hydride when it is -1
- Nitrogen is +4 in nitrogen dioxide, +5 in nitric acid and +3 in nitrous acid
 - It has been reduced from +4 in nitrogen dioxide to +3 in nitrous acid
 - It has been oxidised from +4 in nitrogen dioxide to +5 in nitric acid
 - Therefore nitrogen has undergone disproportionation

